

The Attention Market: Visualising the Hidden Systems of Data Capitalism

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Abstract. This thesis examines how interactive data visualisation can make the hidden mechanisms of social media data collection legible to ordinary users. Drawing on Zuboff’s surveillance capitalism, Lessig’s code-as-law argument, and the attention economy literature, it develops The Attention Market — a browser-based, stock-market-styled visualisation that renders aggregated social media data flows as an immersive experience. Where privacy policy and regulation have failed to make these systems felt, this project asks whether design can succeed. The methodology uses p5.js with open-source and synthetic datasets, evaluated through a small user study measuring shifts in awareness.

Keywords: Data capitalism · Surveillance capitalism · Data visualisation · Attention economy · Privacy · Interactive design · p5.js

1 Introduction

Over the past decade, AI has not just changed technology, it has also quietly reshaped the foundations of modern capitalism. The engine driving that shift isn’t code or computing power, it’s largely due to the large accumulation of data.

Back in 2006, British mathematician and entrepreneur Clive Humby put it simply: “Data is the new oil.” The quote carries a double meaning. On one side it signals that raw data, like crude oil, is worth little on its own, but once it has been processed, structured, and put to work it becomes crucial for developing technology in modern society. On the other, just as the parties who own the oil hold the most wealth and power over those who need it, the same logic applies to data in the digital economy.

The AI breakthroughs that dominate headlines today are built on vast volumes of user data [3]. The user here is quite literally anyone using any form of social media or internet platform. These developments would not have been possible without people generating the behavioral data that AI models have been training on — meaning that in many ways, users have been the invisible labor force behind the AI revolution without ever being compensated or even fully aware of their contribution.

Yet most people scrolling through their feeds have little sense of how deeply they are embedded in this system. Every time we tap “Agree” on a platform’s terms of service, we sign over our behavioral data for commercial use, often

without really knowing it. Cookies trace every click, every pause, every pattern of attention across the web. This model, broadly referred to as data capitalism, turns human attention and behaviour into tradeable commodities, fuelling the trillion-dollar valuations of a handful of tech companies. Zuboff [11] goes further, arguing that this data does not just hold monetary value but also functions as a mechanism of power, which she terms surveillance capitalism. The profit incentive behind free social media platforms is advertising. Once Big Tech companies like Meta and Google began collecting user data at large scale, they built AI systems capable of predicting future user behaviour, allowing advertisers to target with increasing precision and platforms to generate higher profits. The key problem is not recommendation systems or personalised advertisements on their own, even if everything stems from there. These are surface features that give most users a sense of fair exchange for the free platforms they use. The real problem begins where these economic systems start interfering with social and political ones, when Big Tech companies no longer just predict behaviour but actively modify and manipulate it. People are vaguely aware something is going on, but the mechanism is hidden and robust, built into the very architecture that governs what we can see, do, and think online. As Lessig [6] argues, code is law — the platforms are not neutral spaces but deliberately designed environments that regulate behaviour in their own favour.

Even as academic and public conversation around digital privacy has grown, the research tells a sobering story. Most users consistently skip over privacy policies [5], and many remain unaware of how extensively their data is collected even when platforms claim to be transparent. This is what has come to be known as the privacy paradox — people say they care about their privacy, then do almost nothing to protect it, building a persistent tension between privacy attitudes and market realities [5]. Corporations accumulate granular knowledge about our lives, personalities, and behaviour whilst we largely operate in the dark.

That imbalance is not neutral. It enables behavioural manipulation that has moved well beyond nudging into something closer to systemic engineering. And with the recent political retreat from AI oversight, most visibly in the rollback of Biden-era AI regulation under the Trump administration, the guardrails are getting weaker, not stronger.

This thesis sits at the intersection of those concerns. It examines the mechanics of data capitalism and asks whether art, design, and visualization might offer something that policy and legal frameworks have not — a way to make these invisible systems legible to ordinary people, and to shift the default from passive acceptance to something more like informed resistance.

The central question I am trying to answer is:

How can data visualizations be used to help people better understand the hidden systems of data collection on social media platforms?

Alongside that, I will be working through a set of sub-questions:

- What mechanisms do social media platforms use to collect and process user data?
- How can these mechanisms be represented through interactive visualization design?
- To what extent can visualization influence users’ awareness of data harvesting practices?

2 Literature Review

In *The Question Concerning Technology* [4], Martin Heidegger warns against the enframing nature of modern technology, when everything in the world, including people are reduced to mere resources to be exploited and optimized. He suggests that the data economy reduced humans to data profiles rather than people. Yet Heidegger holds space for resistance through art, suggesting that the authentic human expression might resist the reductionist view through awareness, regulation and alternative design.

The most direct problem this project seeks to visualize is what lies underneath is the deliberate invisibility of these systems. Heidegger warned us about the enframing nature of technology, but writing before the rise of platform capitalism he could not have anticipated how precisely that condition would be engineered. What he did identify was its essential logic that it only works because people do not feel it or see it. Transparency would invite resistance. Zuboff [11] picks this up in a more contemporary context, showing how platforms that present themselves as offering free services and connection are running something quite different underneath. Every interaction is turned into behavioural data, reframed to the user as personalisation, as if the algorithm exists to serve you rather than extract from you. What makes her argument particularly unsettling is the shift from prediction to modification where these systems do not just anticipate your future behaviour, but they are designed to nudge it. The advertisement companies are not buying your data, rather than buying the certainty over what you will do next, as well as the ability to influence it. The consequences stretch well beyond targeted ads, as the role of behavioural data in political campaigning has shown, most visibly in the case of Cambridge Analytica [1,11]. The real question this raises is not whether you have something to hide, but whether someone has the tools to influence you. The system works because it does not feel like a system. That invisibility is a deliberate design of platform capitalism not an inevitable consequence of technological enframing.

The designs in place on social media platforms today are built intentionally to attract your attention. In a world that’s filled with information the recourse companies fight for becomes the users attention since that is what generates data that could be further sold and used as a commodity [7,10]. For example, introduced in the recent decade features like infinite scrolling and autoplay leave the user hooked for hours without any control over when the media stops playing. Lessig’s core argument in *Code and Other Laws of Cyberspace* [6] is that code is law, it regulates our behaviour, explaining how code is non-negotiable, allowing

it to operate invisibly and by default. Only the companies that design this architecture have the power to govern over what happens internally with users and their information. Thus, granting them with the power and tools they wish to use, letting them design the platforms to do the political and commercial work for them, by essentially legislating user behaviour in their own favour, outside of any democratic process.

When the internet is not governed by any specific unified delegated parties, thus, the structural power behind social media mechanism is determined by the media companies themselves. Varoufakis [9] argues that they have intentionally built feudal structures to exploit for their own benefit. As we surf the internet on free media platforms the invisible architecture behind it greatly benefit financially by extracting information when people are using them. The asymmetry between users and the platforms is difficult to challenge since even if many people are aware of it, the design that Lessig [6] argues governs our behaviour and it becomes increasingly harder to escape the paradox of their power over us.

There is various information sources available on this issue as users express their concerns, but often fail to act on them suggesting that there is something deeper going on that prevents them from taking action even when they are fully aware. This presents the privacy paradox as it creates a gap between identified values and actual behaviour [5]. The legal systems in place such as the GDPR [2] enforce companies to put privacy laws into place especially in Europe. However, when big companies such as Meta and Google, get penalized there exists a cap on the fraction of revenue they can take, which to such huge profit companies is close to merely nothing, which creates an unfair distribution of power for big tech companies vs small competitors and the regulatory body. This kind of policy response as well as many others simulate the illusion of transparency, whilst in reality they fall short. This gap is precisely where my project comes into place, instead of exhausting users with endless agreement paperwork framed in a very legal language, I will try to create a perceptual experience and make the reality of the system feel like an experience.

A research paper by Megan L. Smith and Yujie Gao, titled “All The Stars We Cannot See: A Deep Look into Real-Time Satellite Traffic” [8], has strongly inspired the direction of this project. Their work combines technical data processing with visual design to make an otherwise invisible issue—satellite congestion caused by Starlink—accessible to a broader audience. While Starlink, a project associated with Elon Musk, is often framed as a scientific and technological advancement, the increasing number of satellites contributes to the pollution of the night sky, creating artificial light patterns that are not immediately visible or understood by the general public.

What is particularly compelling about this work is the way the authors translate complex, real-time data into an interactive visualization tool that allows users to perceive the scale and impact of satellite activity. By doing so, they bridge the gap between abstract data and public awareness, enabling users to critically engage with an issue that would otherwise remain hidden. This approach strongly aligns with the goals of my thesis. Similarly, I aim to explore

how data visualization and interactive design can be used to expose less visible forms of “digital pollution,” specifically in the context of social media and data harvesting. By transforming abstract data processes into accessible and engaging visual representations, I hope to create a tool that raises awareness and encourages users to reflect on their own participation within these systems.

3 Methodology

The goal of this project is to build an algorithm that tracks real-time data traffic across social media platforms and translates it into a visual interface, creating an immersive experience that helps users better understand data mining. Ideally, using p5.js or TouchDesigner, the algorithm would pull from API-driven real-time data and turn it into an interactive visualization that builds and shifts over time, reflecting the continuous flow of data moving through these systems.

The artifact itself would look and feel like a stock market interface, built into a webpage. Intentionally creating the urgency, the volatility, the sense of constant competition. Following Lessig’s [6] argument that design is never neutral, the visual language here is meant to expose rather than glorify. The project would be called The Attention Market. It draws a direct line between the financial logic of social media and how we engage with it, grounding the piece in the ideas of data capitalism and the attention economy. It is essentially trying to make visible what Zuboff [11] argues theoretically, that our attention is the raw material being extracted.

Getting hold of large-scale real-time social media data is where things get complicated. Personal data is easy enough to access, but this project is not about individual behaviour. The aim is to reflect collective patterns, which means working with aggregated, anonymous, non-user-specific data that sits within European privacy regulations. That boundary is non-negotiable for this project. Meta’s Content Library API is one option, offering aggregated engagement data with timestamps and geographic detail, but access requires an application that might not clear within the project timeline. Third-party providers exist too, though most are commercially driven and sell data to advertisers, which puts them out of reach financially. Scraping is worth mentioning, and some academic exemptions do apply, but the legal grey areas and potential delays make it unreliable as a primary method. What does exist, thankfully, is a body of open-source datasets left behind by researchers who have gone through these processes before.

Given all of that, the most practical option is probably building on those open-source datasets and using them as the foundation for synthetic data generation that simulates continuity over time. It is the most realistic option, though I want to be honest about what that means if users know the data is synthetic then the emotional weight of the piece could change the impact. The rawness of it might somewhat disappear. If that ends up being the case, the project still works as a starting point for a bigger conversation, one that could grow through future collaboration with institutions that have the access and infrastructure to work with real data properly.

For the actual build, p5.js makes the most sense. It comes from a creative coding background, which fits the nature of this project better than something like TouchDesigner, which is more visually powerful but harder to control for this kind of metaphor-driven work. D3.js and Chart.js are the obvious alternatives but they lean too conventional for what this piece is trying to do. p5.js sits in the right middle ground, and the fact that it runs in a browser makes user testing straightforward. The evaluation would run as a small user study, looking at emotional response and how well people actually grasp what they are seeing, and pointing them toward concrete ways they could protect their own data. The measure of success here is whether it genuinely shifts how someone thinks about their relationship with social media, not just in the abstract but personally.

The scope is deliberately managed. The priority is getting the visualization right, not solving the data access problem, which carries legal and ethical dimensions that go beyond this project. The data side will be covered thoroughly in the thesis writing itself, and with more time and the right partnerships, the full version of this project is something worth returning to. The core aim of this project is to transform abstract data traffic on social media platforms into a visually engaging experience that enhances the users understanding of data mining. The idea is to design an algorithm that tracks real-time data streams and outputs an immersive interface, with the help of tools like p5.js transform it from abstract in order to design a visual interface that delivers an immersive experience for the user, enabling a better understanding of data mining.

The study measures four dimensions: awareness shift, emotional response, behavioural intention, and the gap between desired and actual social media use. Given the project's grounding in the attention economy, the study will also capture participants' self-reported social media habits. Before the study, participants will be asked to estimate their average daily screen time on social media and evaluate how much they want to use it versus how much they actually use it on a scale of 1 to 10. The gap between desired and actual usage directly reflects the behavioural lock-in that Simon [7] and Wu [10] describe. After the visualisation, the same question about desired usage is repeated, to test whether exposure shifts their perceived relationship with the platforms. This won't capture real behavioural change, which would require a longitudinal follow-up beyond the scope of this project, but it gives a rough proxy for whether the piece disrupts the normalised acceptance of excessive platform use.

4 Planning

The project runs from April through to July 2026. A lot of draft writing and research has already been completed. Additionally, if there are any delays then with a late extension the project can be submitted no later than 31st of July 2026.

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